Predicting Sleep Quality from Wearable Multisensor Data

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Introduction

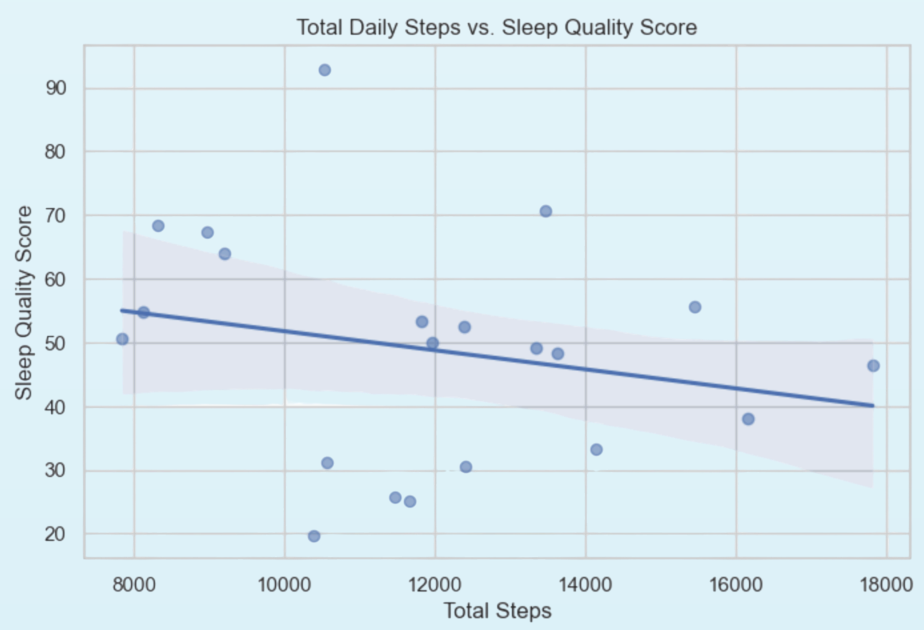
Sleep quality is a critical component of overall human health, yet continuous and objective monitoring remains difficult. Traditional clinical methods such as polysomnography are highly accurate, but they are invasive, expensive, and usually restricted to laboratory settings. Wearable technology provides a more accessible alternative for tracking physiological and behavioral metrics in daily life, but converting raw wearable data into an interpretable sleep outcome remains a challenging data science problem. This project addresses that problem by building a data-driven model that uses daytime wearable and demographic features to predict an engineered continuous Sleep Quality Score.

We use the Multilevel Monitoring of Activity and Sleep in Healthy People (MMASH) dataset, which includes participant sleep records, actigraphy data, heart rate data, beat-to-beat interval data, activity logs, questionnaire scores, and demographic information. After preprocessing and dropping one participant with incomplete required variables, the final cleaned dataset contains n = 21 observations and p = 6 predictors: age, BMI, average heart rate, total daily steps, mean vector magnitude of movement, and number of heavy activity events. Because the dataset does not provide a single clinical ground-truth sleep quality label, we construct a composite Sleep Quality Score on a 0 to 100 scale, derived from total sleep time, sleep efficiency, wake after sleep onset (WASO), number of awakenings, and sleep fragmentation index, where higher values indicate better sleep quality.

Hypothesis

We hypothesize that participants with higher daytime physical activity will have higher Sleep Quality Scores. This is motivated by prior evidence that physical activity promotes deeper and more consolidated sleep. Specifically, we expect total_steps, mean_vector_mag, and heavy_activity_events to be positively associated with the Sleep Quality Score.

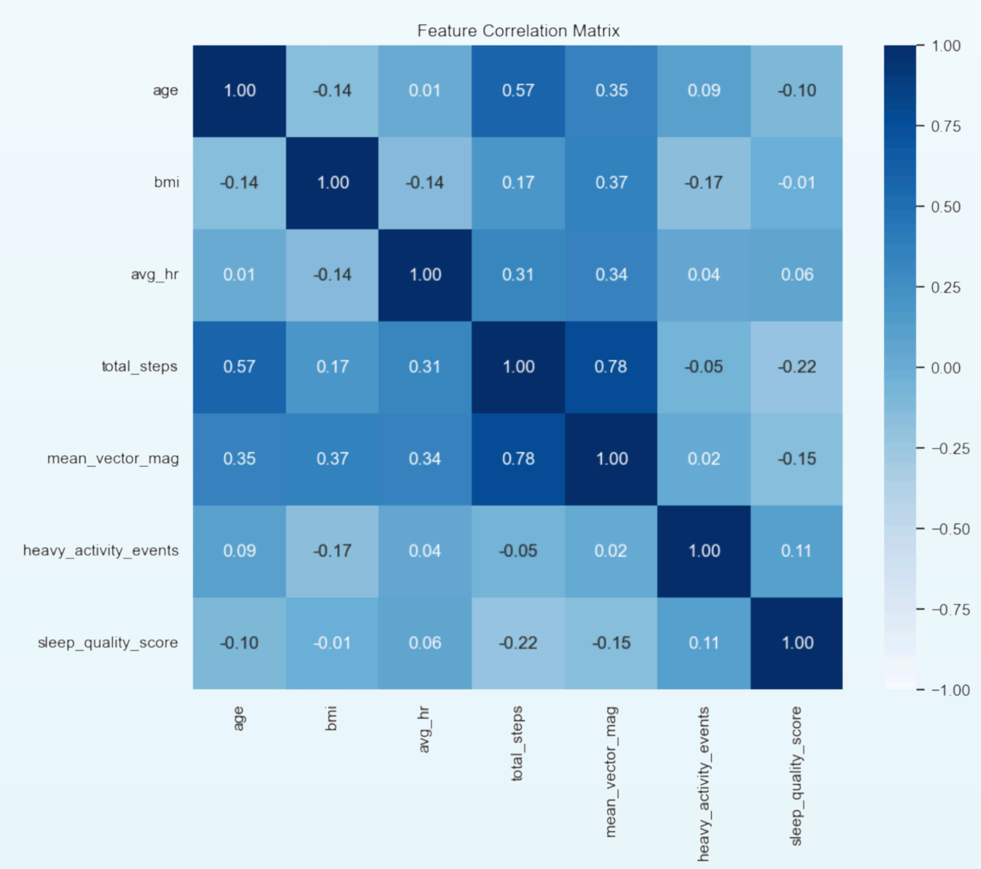
Null hypothesis: daytime physical activity is not associated with the Sleep Quality Score. This hypothesis is evaluated using both prediction performance and coefficient direction. Prediction performance is measured with cross-validated mean squared error (MSE), and coefficient direction is used only for exploratory interpretation because the sample size is small.



Methodology

Data Processing and Target Variable Construction:
The MMASH dataset is structured across 22 distinct participant folders, each containing individual CSV files spanning a continuous 24-hour window across two calendar days. Our preprocessing script iteratively accesses each folder and chronologically merges continuous biosignals to capture a single consolidated wake-sleep cycle per participant.

We aggregate second-by-second wearable data and categorical activity logs into daytime summary statistics. Because the dataset does not provide a ground-truth sleep score, we compute a composite Sleep Quality Score scaled from 0 to 100, representing a weighted linear combination of positive restorative factors (Total Sleep Time, Latency Efficiency) penalized by sleep disruptions (Wake After Sleep Onset, Sleep Fragmentation Index, and Number of Awakenings).



Model Estimation and Validation Strategy:
We use 5-fold cross-validation on shuffled observations to ensure generalization and remove subject-ordering bias. In each iteration, the model trains on 80 percent of the data and tests on the remaining 20 percent, with final performance reported as the average MSE across all five folds. Feature standardization is performed inside the pipeline to prevent data leakage, and Lasso hyperparameter tuning is handled within cross-validation via GridSearchCV over alpha values from 0.001 to 1.0. We also include a naive mean baseline using DummyRegressor, which predicts the training-set mean for each test participant, to confirm that our regression models improve on a constant prediction.

Feature	Lasso coefficient
age	0.000
bmi	0.000
avg_hr	1.077
total_steps	-3.221
mean_vector_mag	0.000
heavy_activity_events	0.744

Results

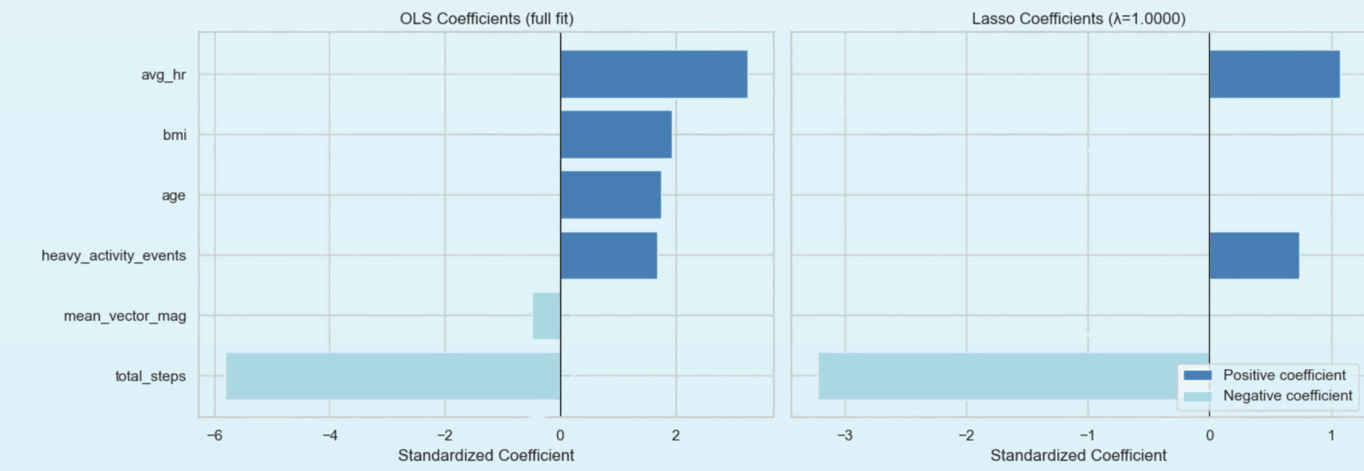
Model	Mean CV MSE	CV MSE Std	RMSE
OLS	454.45	296.79	21.32
Lasso	361.38	275.90	19.01
Naive mean baseline	293.83	211.14	17.14

After dropping one participant with incomplete data, the final dataset contained n = 21 observations. The engineered Sleep Quality Score ranged from 19.75 to 92.93 (mean = 48.94, SD = 17.52), forming a roughly normal distribution centered near the midpoint of the 0 to 100 scale. This spread is favorable for regression, as it provides meaningful variation in the target variable for the model to detect.

The baseline OLS model produced a mean 5-fold CV MSE of 454.45 (SD = 296.79), corresponding to an RMSE of approximately 21.3 sleep score points. The wide standard deviation across folds (ranging from 146.13 to 854.87) reflects substantial instability, which is consistent with the high predictor-to-observation ratio (p/n). Fitting six predictors on only 21 observations leaves OLS highly susceptible to overfitting.

The Lasso model produced a mean 5-fold CV MSE of 361.38 (SD = 275.90), corresponding to an RMSE of approximately 19.01 points, a 20.5% reduction relative to OLS. Lasso was selected as the preferred model based on this lower CV MSE. The improvement is attributable to Lasso eliminating three of six predictors entirely (age, BMI, and mean_vector_mag), removing features that were adding noise rather than signal in a small sample. This is the expected behavior when multicollinearity and a high p/n ratio are present.

Given our small sample size (n = 21) relative to the number of predictors (p = 6), highly flexible models are prone to overfitting. The cross-validation results reflect this complexity. The flexible OLS model exhibited the highest error (Mean CV MSE: 454.45), indicating it memorized the training folds but generalized poorly. Lasso regression successfully reduced this complexity by shrinking redundant coefficients, improving the Mean CV MSE to 361.38. However, both regression models were outperformed by a naive baseline. The variance of the target variable is approximately 307 (SD^2≈ 17.52^2), and simply predicting the mean score for every participant yields an RMSE of 17.14. The flexible models produced higher RMSEs (19.01 and 21.32), introducing more noise than predictive signal. Neither model can be considered a reliable predictor of sleep quality from these daytime features alone.



Conclusion

The original hypothesis predicted that higher daytime physical activity would be associated with better sleep quality. The results do not provide strong support for this hypothesis. Lasso performed better than OLS, which supports the idea that regularization is helpful when predictors are correlated and the sample size is small. However, the naive mean baseline still produced the lowest cross-validated error, meaning the regression models did not improve on a simple constant prediction.

The most defensible conclusion is that the engineered Sleep Quality Score is not well predicted by these six daytime features alone in this small dataset. The weak performance may be due to the small number of participants, noise introduced by constructing a composite sleep score, individual differences in sleep physiology, and the possibility that daytime activity features alone are insufficient for predicting sleep quality.

The coefficient results should therefore be treated as exploratory rather than confirmatory. The negative coefficient for total_steps does not prove that walking more causes worse sleep. It only shows that, in this small sample and engineered-score framework, higher step count was associated with lower predicted sleep quality after standardization and regularization.

Key Takeaways

More daytime activity does not automatically mean better sleep. In our sample, higher step count was actually slightly associated with lower sleep quality, so the relationship between exercise and sleep may be more complicated than expected.
Sleep quality is influenced by multiple factors together, not just one variable like steps or heart rate.

Very intense or high-activity days may not always improve sleep. They could sometimes lead to fatigue, stress, or physical arousal that makes sleep less restful.
Average heart rate and heavy activity showed some positive relationship with sleep quality, but the evidence was weak, so we cannot say they clearly cause better sleep. Wearable data can be useful for studying sleep, but simple daytime features alone may not be enough to predict sleep quality accurately.

Sleep quality may depend more on things we did not measure, such as stress, caffeine, screen time, bedtime routine, mental health, diet, or sleep environment.

References

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